Shaswat Pathak



Aspiring Physicist & Researcher — M.S. (Research) in Physics (Theoretical and Computational Physics)

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About me

I am a Master's student in Theoretical and Computational Physics at Universiti Sains Malaysia, working under Prof. Yoon Tiem Leong on inverse problems and quantum-corrected light-matter interactions in plasmonics. My research interests include theoretical quantum physics, condensed matter physics, quantum materials, and their applications. I am also experienced in first-principles calculations (DFT/TDDFT) of phonon dynamics, electronic properties, and molecular interactions with 2D materials, as well as machine learning and numerical simulations to develop interpretable, physics-informed approaches to quantum many-body and multiscale problems.

Education

Universiti Sains Malaysia, Penang, Malaysia Nov 2025 – Ongoing

M.S. (Research) – *Theoretical and Computational Physics* (**Specialization:** Condensed Matter Physics)

- Coursework completed, thesis in progress; thesis submission November 2026. – Grade-Sheet (Sem-1)
- **Thesis:** PINNs for Inverse Retrieval of Nonlocal Parameters in Quantum Plasmonic Nanostructures. (Proposal, Slides)

SRM Institute of Science and Technology, TN, India July 2020 – May 2024

B.Tech. – *Mechanical Engineering*

- **CGPA:** 9.40/10.00 (SGPA 10.00/10.00 in final two semesters) – Transcript, Degree
- **Relevant Coursework:** Electromagnetic Theory, Wave Optics, Quantum Mechanics, Thermodynamics, Fluid Mechanics, Advanced Calculus, Linear Algebra, Engineering Mechanics, Heat and Mass Transfer.
- **Undergraduate Theses:** DNN-accelerated GPU cooling system design using CFD and Particle Swarm Optimization (Thesis); CFD analysis of two-phase heat and mass transfer in a modified double-slope solar still (Thesis)

Publications

Physics & Computational Materials Science

- **Pathak, S.**, Dwivedi, V., Linda, A., Bhowmick, S. (2026). *Explainable Learning of Debye Temperature for Physics-Guided Materials Discovery.* Materials Today Physics – **Accepted**. Manuscript
- Linda, A., Akhtar, M.F., **Pathak, S.**, Bhowmick, S. (2024). *Accelerating the prediction of stacking fault energy by combining ab initio calculations and machine learning.* Phys. Rev. B, 109, 214102. doi:10.1103/PhysRevB.109.214102

Mechanical & Thermal Engineering

- **Pathak, S.**, Luitel, A., Pasha, A.A., Reddy, D.S.K., Algarni, S., Alqahtani, T., Irshad, K. (2025). *ML-augmented CFD framework for heat-sink and fan-speed optimization in gaming workstations.* Int. J. Mech. Sci., 110752. doi:10.1016/j.ijmecsci.2025.110752
- **Pathak, S.**, Jeyaraj, T., Kumar, P. (2025). *CFD and Experimental analysis of a double-slope solar still with channel integration.* Appl. Therm. Eng., 258, 124538. doi:10.1016/j.applthermaleng.2024.124538
- **Pathak, S.**, Kumar, P., Jeyaraj, T. (2025). *Enhancing energy efficiency in a modified double-slope solar still: channel optimization and salt concentration effects.* J. Therm. Anal. Calorim. doi:10.1007/s10973-025-15107-6
- Jeyaraj, T., Kumar, P., **Pathak, S.** (2025). *Performance enhancement of a double-slope solar still using integrated channel shape variations.* Renew. Sustain. Energy Rev., 209, 115106. doi:10.1016/j.rser.2024.115106
- Jeyaraj, T., Kumar, P., **Pathak, S.** (2024). *Experimental and numerical modeling of a double-slope solar still with trapezoidal channel preheating.* Appl. Therm. Eng. doi:10.1016/j.applthermaleng.2024.123757

Current Research and Additional Projects

- **Physics-Informed Modeling of Quantum-Corrected Nonlocal Plasmonic Response in Gold Nanoparticles** – Coupling a quantum-corrected hydrodynamic model with the local response approximation of Maxwell's equations for inverse plasmonic parameter retrieval in metallic nanostructures and metamaterials. [Manuscript in Preparation]
- **Uncovering Macroscopic Tuning Pathways for Quantum Lattice Dynamics and Superconductivity with XAI** – Establishing interpretable relationships between macroscopic elastic descriptors, Debye temperature, and superconducting transition temperature using data-driven modelling and symbolic regression. [Manuscript in Preparation]
- **First-Principles Investigation of Molecular Adsorption and Electronic Structure Modulation on γ-Graphyne Nanotube** – Adsorption energy, QTAIM/ELF, and electronic-structure analysis for Ribavirin/Favipiravir delivery. [Manuscript in Preparation]
- **PINNs and VMCNN for Solving Asymmetric Double-Well Schrödinger Equations** – Physics-informed variational MC neural-network solver for analytically intractable time-independent Schrödinger equations with double-well potentials. [GitHub Code]

Work Experience

Senior Project Mechanic – CMS Lab, IIT Kanpur, India (Full-time, Certificate) Apr – Sep 2025

- Developed an XAI framework integrating symbolic regression, latent-space, and critical-regime analyses to uncover governing descriptors of lattice dynamics, derive interpretable scaling laws for Debye temperature and lattice conductivity, and discover advanced functional ceramics for thermoelectric and superconducting applications with persistent quantum vibrational dynamics at high temperatures.

Research Trainee – Center for Advanced computational research, New Delhi, India (PT, Certificates) Jun 2024, Aug 2026

- Performed DFT calculations using Quantum ESPRESSO and Gaussian to investigate electronic structure, stacking-fault energies, and drug–material interactions; designed nanostructures and quantum dots using VESTA and BURAI. Completed hands-on training in TDDFT, harmonic and anharmonic phonon calculations, electronic-property calculations, and elastic-moduli calculations using Quantum ESPRESSO and ThermoPW for computational characterization of materials.

Teaching Assistant, High-Performance Computing – SRMIST, India (Part Time, Certificate) Jan – May 2025

- Assisted with Python-based HPC workflows (NumPy, SciPy, MPI) and numerical methods for PDE-based large-scale CFD problems.

Undergraduate Researcher – Nano and molecular simulation lab, SRMIST, India (PT, Certificate) Nov 2023 – Apr 2025

- DFT-based quantum-chemistry studies (DOS, ELF, FMO) for large-molecule interactions with B-N-Zn-doped γ-graphyne nanotubes.

R&D Intern – Fluidyn Pvt. Ltd., Chennai, India (Part Time, Certificate) Jun – Nov 2024

- Built a Python finite-difference solver for radon advection–diffusion transport and developed ANN- and PSO-based surrogate models for uncertainty-aware radon flux prediction and optimization.

Student Researcher – Fluid Dynamics & Electronic Cooling Labs, SRMIST, India (PT, Certificates) Jun 2022 – May 2024

- Improved solar-still yield (up to 37.64%) by addition of microchannels and CFD-driven design optimization. Developed a hybrid CFD–ANN–PSO framework for high-power workstation thermal model and GPU cooling optimization.

Research Intern – CMS Lab, IIT Kanpur, India (Full Time, Certificates) Jan – Aug 2023

- Developed VMCNN-based PINNs for solving the TISE with asymmetric double-well potentials. Combined DFT, Random Forest feature selection, and cubic regression to derive interpretable scaling laws linking d-band descriptors to stacking-fault energies and shear modulus.

Intern (Nanotech) – Teachnook & Wissenaire, IIT Bhubaneswar, India (Part Time, Certificates) Apr – May 2022

- Studied nanomaterials, nano-biomaterials, & quantum science. Projects: $BaTiO_3$ synthesis and Nanomaterials application in Mech Eng.

Leave of Absence

None (no career breaks or periods of leave).

Conference and Seminar Presentations

- **Deep learning to solve TI-Schrödinger Equation with a Asymmetric Double-Well Potential** – *Annual Research Day*, SRMIST (Mar 2024, Certificate)
- **Improving Cooling Performance of Heat Sink and Heat Pipes of a Workstation with CFD and Optimization Tools** – *International Conf. on Advances in Mechanical Engineering*, Chennai (Mar 2024, Certificate)
- **Comparative Analysis of Regression Models in an Engineering Optimization Problem** – *International Conf. on Mathematical Techniques and Applications*, Chennai (Jan 2024; Certificate)
- **Study of Characteristics of Two-Phase Flow in a Double Slope Solar Still with Channels** – *All India Seminar on Solar Energy for Sustainable Development* (Sep 2023, Certificate)

Awards and Achievements

Prestigious Scholarships and Selections

- **Chevening Scholarship (Final Interview) & Rhodes Oxford Scholarship (2nd Round)**, UK (2024–2025) [Link] – Shortlisted from over 100,000 applicants for academic excellence, leadership, and interdisciplinary research contributions.
- **Selected, MSc Physics at University College London** (QS Ranking #9, 2025) [Link] – Merit-based admission offer.

Research Grants and Awards

- **Large Group Research Grant (RGP2/266/45)**, Deanship of Scientific Research, King Khalid University, Saudi Arabia (2025) – ML-based heat-sink/fan design optimization. [Publication Link]
- **Best Student Researcher Award**, SRMIST (2024) [Certificate] – For interdisciplinary research and undergraduate publications.
- **Best Project & Best Thesis Presentation Awards**, SRMIST (2024) [Certificates] – For B.Tech thesis on an ML-integrated CFD framework for energy-efficient GPU cooling optimization.

Academic Excellence Awards

- **Dean's Award for Academic Performance**, SRMIST (2023) [Certificate] – Top 1% of ME Department, INR 51,250 scholarship.
- **NPTEL Elite Topper**, IIT Kanpur (2022) [Certificate] – All India Rank 1 in "Electronic Properties of the Materials."
- **SAP Selection and Department Rank 1**, SRMIST (2022) [Link] – SAP selection to Umeå University, Sweden. Top ranks in Semesters 4, 7, and 8 at SRMIST.

Skills

- **Theoretical:** Quantum Mechanics, Statistical Mechanics, Condensed Matter Physics, Electronic properties of Materials, Quantum Theory of Lattice Dynamics (Phononics), Plasmonics, Electromagnetic Response of Metals, and Density Functional Theory.
- **Computational Physics and Modelling:** EM scattering & Nonlocal plasmonics; Inverse problems; DFT for electronic-structure, mechanical properties, and lattice-dynamics calculations; CFD for heat–mass transfer and two-phase flow; Monte Carlo, Numerical Methods & PDE/ODE solvers.
- **ML/Data-Driven Modelling :** PINNs; XAI; Interpretable ML (Symbolic regression, PCA/PLS, Critical regime analysis, Monte Carlo Input Perturbations); DNN/MLP, Random Forest, Gaussian process & SVM regression; Particle Swarm Optimization; High-Throughput Database Querying (AFLOWLIB, Materials Project, Phonondb).
- **Softwares/Codes:** MATLAB, Quantum ESPRESSO, VASP, QuantumATK, Gaussian, ThermoPW, ANSYS Fluent/Icepak, VESTA, BURAI, LaTeX.
- **Languages:** ENGLISH (Full Professional Proficiency – IELTS-7/9), HINDI (Native Language), JAPANESE / BAHASA MELAYU / SANSKRIT (Elementary Proficiency, Part of Coursework).

Additional Coursework, Workshops and Webinars

- **Ongoing Courses:** Exploring Quantum Physics (University of Maryland, Coursera); Applied Calculus (Harvard, edX); Nanophotonics & Plasmonics (UT San Antonio, Lecture series by Nicolas Large).
- **Completed Courses:** Electronic Properties of Materials [Link] & Fundamentals of Electronic Materials and Devices [Link] (IIT, NPTEL); Nanotechnology [Link] (IIT Bhubaneswar, Teachnook); Statistical Mechanics (Lecture Series by Leonard Susskind); Python Libraries for Data Science [Link] & Introduction to Data Analytics [Link] (Simplilearn).
- **Workshops:** Quantum machine learning [Link], Observational astronomy using telescope and Computational physics [Link], Visualizing graphs and recursions [Link] (IIT Madras); Space science theory, technology, and applications [Link] (ISRO/IIRS).
- **Attended Webinars:** Seeing the Quantum Nanoworld (Oxford University); Bright Space Vacuum: Quantum Effects in Macroscale (UCL); Designing Quantum Materials with Light (AttoFridays); Quantum Computing Meets Artificial Intelligence (University of Tokyo).

REFERENCES

1. **Prof. Somnath Bhowmick** [Website] – *Professor, Department of Materials Science and Engineering, Indian Institute of Technology (IIT), 208016, Kanpur, India.* E-mail: bsomnath@iitk.ac.in;
2. **Prof. Yoon Tiem Leong** [Website] – *Associate Professor, School of Physics, Universiti Sains Malaysia, 11800 USM, Penang, Malaysia.* E-mail: tlyoon@usm.my;
3. **Dr. Albert Linda** [Website] – *Department of Sustainable Metallurgy and Alloy Design, Max Planck Institute for Sustainable Materials, 40237, Düsseldorf, Germany.* E-mail: a.linda@mpi-susmat.de;

Additional Academic References

1. **Prof. T. Vijayakumar** [Website] – *Associate Professor, Department of Physics and Nanotechnology, SRM Institute of Science and Technology, 603203, Chennai, India.* E-mail: vijayakt1@srmist.edu.in;
2. **Prof. Hariharan R. M.** [Website] – *Associate Professor, Department of Physics and Nanotechnology, SRM Institute of Science and Technology, 603203, Chennai, India.* E-mail: hariharr@srmist.edu.in;
3. **Prof. Siva Krishna Reddy** [Website] – *Research Assistant Professor, Department of Mechanical Engineering, SRM Institute of Science and Technology, 603203, Chennai, India.* E-mail: sivakriv@srmist.edu.in;
4. **Prof. Pankaj Kumar** [Website] – *Research Assistant Professor, Department of Mechanical Engineering, SRM Institute of Science and Technology, 603203, Chennai, India.* E-mail: pankajkr@srmist.edu.in;